AI Foundations: VS Code + R Laptop Setup Guide

These instructions cover macOS and Windows. Platform-specific steps are noted where they differ.

1. Download workshop files

- Go to the workshop repository on GitHub (link provided by your instructor)
- Click the green **Code** button, then click **Download ZIP**
- Extract the ZIP file to a location you'll remember (e.g., your Desktop or Documents folder)

> You do not need Git installed. The ZIP download includes everything you need.

2. Install Homebrew (macOS only)

If you already have Homebrew, skip this step. Otherwise, open **Terminal** (`Cmd + Space` , type `Terminal` , press `Enter`) and run:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Follow the prompts. When it finishes, **close and reopen Terminal** so the `brew` command is available.

3. Install R (macOS)

Option A: Homebrew (if you completed Step 2)

```
brew install r
```

Option B: CRAN installer (recommended for most users)

- Download the latest `.pkg` from https://cran.r-project.org/bin/macosx/
- Open the installer and follow the prompts

3. Install R (Windows)

- Download the latest `.exe` from https://cran.r-project.org/bin/windows/base/
- Run the installer and follow the prompts

4. Install Node.js (macOS)

Claude Code requires Node.js version 18 or later.

Option A: Homebrew (if you completed Step 2)

```
brew install node
```

Option B: Installer

- Download the LTS installer from https://nodejs.org/

Verify by running `node --version` in Terminal. You should see `v22.x.x` or similar.

4. Install Node.js (Windows)

Claude Code requires Node.js version 18 or later.

- Download the LTS installer from https://nodejs.org/
- Run the installer and follow the prompts (accept the defaults)

Verify by opening **Command Prompt** and running:

```
node --version
```

5. Request an API key

Email Nicholas Warren at DISSC (dissc@yale.edu) to request an AWS Bedrock API key for Claude. **Request this before the workshop;** you'll need the key in Step 6. This key will expire after the workshop. For long-term access, contact dissc@yale.edu.

6. Configure AWS Bedrock credentials (macOS)

In Terminal, run `nano ~/.zprofile` and add these three lines, replacing `TOKEN_HERE` with your API key from Step 5:

```
export AWS_REGION="us-east-1"
export AWS_BEARER_TOKEN_BEDROCK="TOKEN_HERE"
export CLAUDE_CODE_USE_BEDROCK=1
```

Save with `Ctrl + O`, press `Enter`. Exit with `Ctrl + X`.

6. Configure AWS Bedrock credentials (macOS, continued)

Back in Terminal, load the new settings into your current session:

```
source ~/.zprofile
```

> If your Mac uses `bash` instead of `zsh` (check with `echo $SHELL`), use `~/.bash_profile` instead.

> **Treat your API key like a password.** Do not share it or commit it to version control.

6. Configure AWS Bedrock credentials (Windows)

Click the Start menu, type `environment variables`, and select **Edit environment variables for your account**. Under **User variables**, click **New** to add each:

Variable	Value
`AWS_REGION`	`us-east-1`
`AWS_BEARER_TOKEN_BEDROCK`	your API key
`CLAUDE_CODE_USE_BEDROCK`	`1`

After saving, restart any open terminals or applications.

7. Install Claude Code

Open a **new** terminal window so it picks up the credentials from Step 6, then run:

```
npm install -g @anthropic-ai/claude-code
```

Verify the installation:

```
claude --version
```

> If you get a permissions error on macOS, run `sudo npm install -g @anthropic-ai/claude-code` instead.

8. Install Quarto (macOS)

```
brew install --cask quarto
```

If you don't have Homebrew, download the installer from https://quarto.org/docs/get-started/

8. Install Quarto (Windows)

- Download the installer from https://quarto.org/docs/get-started/
- Run the installer and follow the prompts

9. Install VS Code

Visual Studio Code (VS Code) is a free, open-source code editor made by Microsoft. It supports R, Python, and many other languages through extensions. We use it in this workshop as your main working environment.

- Download and install from https://code.visualstudio.com/

10. Install VS Code extensions

- Open VS Code
- Click the **Extensions** icon in the left sidebar (`Cmd + Shift + X` / `Ctrl + Shift + X`)
- Search for and install each of these:
  - **R** by REditorSupport
  - **Quarto** by Quarto

11. Configure VS Code

> If VS Code was already open during Step 6, quit and reopen it so it picks up the new credentials.

- `File > Open Folder` and select the `vs-code-and-r` folder from Step 1
- When prompted, click `Yes, I trust the authors`

11. Configure VS Code (continued)

- Open the Command Palette (`Cmd + Shift + P` / `Ctrl + Shift + P`)
- Type `User Settings JSON` and select `Preferences: Open User Settings (JSON)`
- If the file is empty or contains only `{}` , replace everything with the contents of `files/machine-setup/vscode/settings.json` and save
- If you already have settings, add the keys from `settings.json` into your existing JSON

12. Start an R session

- In VS Code's left sidebar, click the top icon (two overlapping pages) to open the **Explorer**, which shows your project files
- Navigate to `files > code` and click `example.qmd` to open it
- Click `R: (not attached)` in the bottom status bar, or run any code chunk — this will start the R session
- If prompted, select the R installation from Step 3
- An **R Terminal** will appear in the bottom panel with an `>` prompt

13. Install R packages

In the R Terminal, run:

```
install.packages(c("tidyverse", "languageserver"))
```

- `tidyverse`: the workshop's dependencies (also installs `broom` and `scales`)
- `languageserver`: powers code completion, go-to-definition, and diagnostics in VS Code

14. Verify your setup

R: In the R Terminal, run `library(tidyverse)`. You should see a startup message listing attached packages.

Claude Code: Open a terminal in VS Code (`Ctrl + ` `) and run:

```
claude "What is 2 + 2?"
```

You should see Claude respond with an answer.